

FI 362 — ADVANCED FINANCIAL ECONOMETRICS

Problem Set 1 | Chapter 2: Linear Time Series Analysis

In-Class Seminar

Mehman Ismayilli

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Instructions. Six technical problems, roughly 10 min each. A calculator is enough — no computer needed. Each problem carries a boxed hint with the formulas you need. Keep three significant figures. Constants: $\chi_{3,0.05}^2 = 7.815$, $\chi_{10,0.05}^2 = 18.31$, $\chi_{12,0.05}^2 = 21.03$; Dickey–Fuller 5% critical values -2.86 (with drift) and -1.94 (no drift).

Problem 1 — Stationarity and the random walk

Splitting the daily S&P 500 sample in half gives:

	first-half mean	second-half mean	first-half sd	second-half sd
log price	7.08	8.06	0.162	0.417
log return	0.0000	0.0005	0.0134	0.0108

- (a) Compute the percentage change in the mean and the ratio (2nd-half sd)/(1st-half sd) for *each* row. State which weak-stationarity condition each series satisfies or violates.
- (b) Model the log price as a random walk with drift $p_t = \mu + p_{t-1} + a_t$, with p_0 fixed and $a_t \sim \text{WN}(0, \sigma_a^2)$. Derive $\mathbb{E}[p_t]$ and $\text{Var}(p_t)$ as explicit functions of t . Are they constant?
- (c) Let $r_t \equiv \Delta p_t = p_t - p_{t-1}$. Compute $\mathbb{E}[r_t]$, $\text{Var}(r_t)$ and the autocovariance γ_ℓ for $\ell \geq 1$, and conclude whether r_t is stationary. What is r_t called?

Hint / formulas. Weak stationarity: (i) $\mathbb{E}[r_t] = \mu$ constant; (ii) $\gamma_\ell = \text{Cov}(r_t, r_{t-\ell})$ depends only on ℓ . Recursion gives $p_t = p_0 + \mu t + \sum_{i=1}^t a_i$. For uncorrelated shocks, $\text{Var}(\sum_{i=1}^t a_i) = t \sigma_a^2$.

Problem 2 — ACF, Ljung–Box, and Box–Pierce

- (a) For the daily S&P ($T = 6640$), $\hat{\rho}_1 = -0.099$. Compute the 5% band $\pm 1.96/\sqrt{T}$ and decide whether $\hat{\rho}_1$ is significant.
- (b) A series of $T = 100$ returns has $\hat{\rho}_1 = 0.25$, $\hat{\rho}_2 = -0.10$, $\hat{\rho}_3 = 0.05$. Compute the Ljung–Box $Q(3)$ and compare with $\chi_{3,0.05}^2 = 7.815$.
- (c) For the same series compute the Box–Pierce statistic $Q^*(3) = T \sum_{\ell=1}^3 \hat{\rho}_\ell^2$. By what percentage does the Ljung–Box refinement raise the statistic here, and why does that correction matter more when T is small?

Hint / formulas. Ljung–Box $Q(m) = T(T+2) \sum_{\ell=1}^m \frac{\hat{\rho}_\ell^2}{T-\ell}$; Box–Pierce $Q^*(m) = T \sum_{\ell=1}^m \hat{\rho}_\ell^2$. Both are χ_m^2 under H_0 : no autocorrelation. The two agree as $T \rightarrow \infty$ because $\frac{T+2}{T-\ell} \rightarrow 1$.

Problem 3 — Estimating and forecasting an AR(1)

Consider the stationary AR(1)

$$r_t = 0.5 + 0.4r_{t-1} + a_t, \quad a_t \sim \text{WN}(0, \sigma_a^2 = 4), \quad r_T = 3.$$

- Find the mean μ and the unconditional variance $\text{Var}(r_t)$.
- Give ρ_ℓ as a function of ℓ , and evaluate ρ_1, ρ_2 .
- Compute the point forecasts $\hat{r}_T(1), \hat{r}_T(2), \hat{r}_T(3)$.
- Compute the forecast-error variances $\text{Var}[e_T(1)], \text{Var}[e_T(2)], \text{Var}[e_T(3)]$ and give their limit as $\ell \rightarrow \infty$.
- Using the closed form $\hat{r}_T(\ell) = \mu + \phi_1^\ell(r_T - \mu)$, reproduce $\hat{r}_T(2)$ and report the one-step R^2 .

Hint / formulas. $\mu = \frac{\phi_0}{1-\phi_1}$, $\text{Var}(r_t) = \frac{\sigma_a^2}{1-\phi_1^2}$, $\rho_\ell = \phi_1^\ell$. Recursion $\hat{r}_T(\ell) = \phi_0 + \phi_1 \hat{r}_T(\ell-1)$; error variance $\text{Var}[e_T(\ell)] = \sigma_a^2 \sum_{j=0}^{\ell-1} \phi_1^{2j}$; one-step $R^2 = \phi_1^2$.

Problem 4 — Order selection: AIC vs BIC thresholds

With $\text{AIC} = -2 \ln L + 2k$ and $\text{BIC} = -2 \ln L + (\ln T)k$, a sample of $T = 200$ ($\ln T = 5.298$) gives:

	$\ln L$	k
Model A	520	3
Model B	523	5

- Compute AIC for both models; which is preferred?
- Compute BIC for both models; which is preferred?
- Let $\Delta \ln L$ be the log-likelihood gain of B over A (here = 3) and $\Delta k = 2$. Find the *break-even* $\Delta \ln L$ at which (i) AIC and (ii) BIC would be exactly indifferent between A and B. Show the actual value 3 sits between them, and use that to explain the split verdict.

Hint / formulas. Model B beats A under a criterion iff its value is lower. Indifference means the criterion difference is zero: AIC ties when $2 \Delta \ln L = 2 \Delta k$; BIC ties when $2 \Delta \ln L = (\ln T) \Delta k$. Lower value = better.

Problem 5 — MA(1): moments, the $|\rho_1| \leq \frac{1}{2}$ ceiling, non-uniqueness

Take the MA(1) (Tsay sign convention) $r_t = a_t - \theta a_{t-1}$ with $\theta = 0.8$, $\sigma_a^2 = 1$.

- (a) Compute $\mathbb{E}[r_t]$, $\text{Var}(r_t)$, ρ_1 and ρ_2 .
- (b) Prove that for *any* MA(1), $|\rho_1| \leq \frac{1}{2}$, and find the θ that attains the bound.
- (c) Given the last reconstructed shock $a_T = 0.5$, compute $\hat{r}_T(1)$, $\hat{r}_T(2)$ and their error variances.
- (d) Compute ρ_1 for the “twin” parameter $\theta' = 1/\theta = 1.25$. What do you find, which of θ, θ' is invertible, and which one does estimation return?

Hint / formulas. MA(1): $\text{Var}(r_t) = (1 + \theta^2)\sigma_a^2$, $\rho_1 = \frac{-\theta}{1 + \theta^2}$, $\rho_\ell = 0$ ($\ell \geq 2$). For (b) maximise $f(\theta) = \frac{|\theta|}{1 + \theta^2}$ via $f'(\theta) = \frac{1 - \theta^2}{(1 + \theta^2)^2}$. Forecast: $\hat{r}_T(1) = -\theta a_T$, $\hat{r}_T(\ell) = \mu$ ($\ell \geq 2$); $\text{Var}[e_T(1)] = \sigma_a^2$, $\text{Var}[e_T(\ell \geq 2)] = (1 + \theta^2)\sigma_a^2$. Invertible iff $|\theta| < 1$.

Problem 6 — ADF test, spurious regression, HAC, long memory

- (a) The ADF regression $\Delta y_t = c + \beta y_{t-1} + \sum_i \gamma_i \Delta y_{t-i} + a_t$ tests $H_0 : \beta = 0$ (a unit root). For the log *price* you obtain $\hat{\beta} = 0.0006$ (se 0.00058); for the log *return*, $\hat{\beta} = -1.02$ (se 0.041). Compute both ADF t -statistics, apply the stated DF critical values, and classify each series as $I(1)$ or $I(0)$.
- (b) Regressing the DAX level on the S&P level gives slope $\hat{\beta} = 3.241$ with OLS se 0.0097. Compute the OLS t . The HAC (Newey–West) se is $4.5\times$ the OLS one — compute the HAC se and the HAC t .
- (c) The regression residuals have $\hat{\rho}_1 = 0.995$. State whether OLS inference is valid, name the phenomenon, and give the correct remedy (be specific about what transformation you would regress instead, and why it works).
- (d) Long-memory ACFs decay hyperbolically, $\rho_\ell \propto \ell^{2d-1}$. Using $\hat{d}_{|r|} = 0.43$, compute the factor ℓ^{2d-1} by which the autocorrelation of $|r_t|$ falls from lag 1 to lag 100. Contrast with an AR(1) short-memory series with $\phi = 0.43$ at lag 100. What does the contrast say, and which chapter-3 model class targets it?

Hint / formulas. (a) ADF $t = \hat{\beta}/\text{se}(\hat{\beta})$; reject the unit root if $t < \text{DF critical value}$. Under a unit root the statistic is *not* normal — use DF values, not 1.96. (b) $t = \hat{\beta}/\text{se}$; HAC keeps $\hat{\beta}$, only rescales the se. (d) $100^{2d-1} = 100^{-0.14}$; AR(1) decays geometrically as ϕ^ℓ . Stationary long memory needs $0 < d < \frac{1}{2}$.